Quantized Assimilation and Larval Feeding Rhythms

A DEB-consistent formulation with empirical grounding

This note formalizes a **bite-quantized** representation of feeding that is *consistent with standard DEB flux notation*: we keep the usual continuous DEB assimilation power $\dot{p}_A$ but represent it as a train of identical intake events ("bites", "sips", "mouthfuls").

Start agnostically: define bite volume and food concentration

Let a single bite remove a **fixed volume** of food medium, denoted by V_{bite} (the *bite volume*, with units $[\text{cm}^3\,\text{bite}^{-1}]$).

Let the food in the medium be described as a DEB substrate concentration X (*food concentration in medium*, $[\text{Cmol}\,\text{cm}^{-3}]$).

Then the amount of food (in C-moles) per bite is

$$\Delta X_{\text{bite}} = X\,V_{\text{bite}} \qquad [\text{Cmol}\,\text{bite}^{-1}]. \tag{1}$$

Reach energy per bite via chemical potential

Let μ_X denote the *chemical potential of food* $[\text{J}\,\text{Cmol}^{-1}]$.

Then the **energy content of one bite** is

$$E_{\text{bite}} = \mu_X\,\Delta X_{\text{bite}} = \mu_X\,X\,V_{\text{bite}} \qquad [\text{J}\,\text{bite}^{-1}]. \tag{2}$$

Define a bite frequency and derive it from DEB assimilation

Let $N(t)$ be the cumulative number of bites. Define the **feeding/bite frequency** as $\dot{N}_{\text{bite}}(t) \equiv \dot{N}(t)$ with units $[\text{bite}\,\text{time}^{-1}]$ (e.g. $[\text{s}^{-1}]$ or $[\text{d}^{-1}]$).

DEB gives a continuous **assimilation power** $\dot{p}_A(t)$ with units $[\text{J}\,\text{time}^{-1}]$.

Quantization assumption: total **ingested food energy** is the sum of bite energies per time:

$$\dot{p}_X(t) = \dot{N}_{\text{bite}}(t)\,E_{\text{bite}}(t). \tag{3}$$

Assimilation is a fixed fraction of ingested food energy:

$$\dot{p}_A(t) = \kappa_X\,\dot{p}_X(t). \tag{4}$$

Therefore, the bite frequency is

$$\boxed{\dot{N}_{\text{bite}}(t) = \frac{\dot{p}_A(t)}{\kappa_X\,E_{\text{bite}}(t)} = \frac{\dot{p}_A(t)}{\kappa_X\,\mu_X\,X(t)\,V_{\text{bite}}(t)}} \qquad [\text{bite}\,\text{time}^{-1}]. \tag{5}$$

Substitute a standard DEB expression for $\dot{p}_A$

A common DEB form is

$$\dot{p}_A(t) = f(t)\,\{\dot{p}_{Am}\}\,S_A(t), \tag{6}$$

where $f(t)$ is the functional response (0–1), $\{\dot{p}_{Am}\}$ is the max surface-area specific assimilation [J time^{-1} cm^{-2}], and $S_A(t)$ is the assimilation surface area [cm^2].

Then

$$\boxed{\dot{N}_{\text{bite}}(t) = \frac{S_A(t)}{V_{\text{bite}}(t)}\,\frac{f(t)\,\{\dot{p}_{Am}\}}{\kappa_X\,\mu_X\,X(t)}}. \tag{7}$$

Break down the functional response and half-saturation

The functional response is

$$f(X) = \frac{X}{K+X}, \quad 0 \le f \le 1, \tag{8}$$

Substituting $f(X)$ into the bite-frequency expression gives

$$\boxed{\dot{N}_{\text{bite}}(t) = \frac{S_A(t)}{V_{\text{bite}}(t)}\,\frac{\{\dot{p}_{Am}\}}{\kappa_X\,\mu_X\,(K+X(t))}} \tag{9}$$

The half-saturation coefficient is

$$K = \frac{\{\dot{J}_{Xm}\}}{\{\dot{F}_m\}}. \tag{10}$$

From the DEB textbook, the maximum ingestion power per area is

$$\{\dot{p}_{Xm}\} = \mu_X\,\{\dot{J}_{Xm}\}, \tag{11}$$

and the corresponding maximum assimilation power is

$$\{\dot{p}_{Am}\} = \kappa_X\,\{\dot{p}_{Xm}\} = \kappa_X\,\mu_X\,\{\dot{J}_{Xm}\}. \tag{12}$$

Therefore,

$$\{\dot{J}_{Xm}\} = \frac{\{\dot{p}_{Xm}\}}{\mu_X} = \frac{\{\dot{p}_{Am}\}}{\kappa_X\,\mu_X}. \tag{13}$$

so

$$K = \frac{\{\dot{J}_{Xm}\}}{\{\dot{F}_m\}} = \frac{\{\dot{p}_{Am}\}}{\kappa_X\,\mu_X\,\{\dot{F}_m\}}. \tag{14}$$

Using Eq. (9) and substituting $K = \{\dot{p}_{Am}\}/(\kappa_X\,\mu_X\,\{\dot{F}_m\})$ gives

$$\boxed{\dot{N}_{\text{bite}}(t) = \frac{S_A(t)}{V_{\text{bite}}(t)}\,\frac{\{\dot{p}_{Am}\}}{\kappa_X\,\mu_X\left(X(t) + \frac{\{\dot{p}_{Am}\}}{\kappa_X\,\mu_X\,\{\dot{F}_m\}}\right)}}. \tag{15}$$

Simplifying the $\kappa_X\,\mu_X$ term yields

$$\boxed{\dot{N}_{\text{bite}}(t) = \frac{S_A(t)}{V_{\text{bite}}(t)}\,\frac{\{\dot{p}_{Am}\}}{\kappa_X\,\mu_X\,X(t) + \frac{\{\dot{p}_{Am}\}}{\{\dot{F}_m\}}}}. \tag{16}$$

The open modeling choice: how should V_{bite} scale with size?

At this point, the only missing ingredient is a **growth-dependent** bite volume $V_{\text{bite}}(t)$, i.e. $V_{\text{bite}}(L)$. A general family with κ_{bite} the coefficient setting the bite magnitude per bite and k the bite-volume scaling exponent:

$$V_{\text{bite}}(L) = \kappa_{bite}\, L^k \qquad [\text{cm}^3\,\text{bite}^{-1}], \tag{17}$$

Two principled options correspond to **area vs volume** scaling, with specific choices of the exponent:

(A) Volume-scaled bites: $k = 3$

$$V_{\text{bite}}(L) = \kappa_{bite,V}\, V(L), \tag{18}$$

where $V(L)$ is structural volume (for an isomorph, typically $V(L) \propto L^3$).

Consequence (for fixed X, f):

$$\dot{N}_{\text{bite}} \propto \frac{L^2}{L^3} \propto L^{-1}. \tag{19}$$

Larger juveniles take fewer bites per time, but each bite is larger.

Interpretation: if you want each bite to represent a roughly fixed fraction of body (structural) volume, choose this volume-scaling option with prefactor $\kappa_{bite,V}$ and exponent $k = 3$.

(B) Area-scaled bites: $k = 2$

$$V_{\text{bite}}(L) = \kappa_{bite,S}\, L^2. \tag{20}$$

Consequence (for fixed X, f):

$$\dot{N}_{\text{bite}} \propto \frac{L^2}{L^2} \propto L^0, \tag{21}$$

so bite frequency tends to remain approximately constant over growth (a "feeding rhythm" interpretation). *Interpretation: if you want a roughly constant feeding rhythm across growth, choose this area-scaling option with prefactor $\kappa_{bite,S}$ and exponent $k = 2$.*

Mouth-hook movement rate across larval development

This section synthesizes **modern neuroethological / circuit-based** and **classical behavioral genetics / developmental** studies relevant to modeling bite / mouth-hook frequency across *Drosophila melanogaster* larval development. Together they delimit what is known, what is measured, and what remains open for DEB-style quantized feeding models.

Modern neuroethological & circuit-based literature

Scope Recent studies focus on the **larval feeding circuit** (brain + subesophageal zone), use **mouth-hook contractions** as a measurable behavioral output, and examine modulation by **hunger, neuromodulators, and developmental state** (see also the review by Miroschnikow et al. [5]). These studies usually work with **late L2 or L3 larvae**, because circuits are mature and behavior is robust.

Key findings Mouth-hook contraction rate emerges as a **stable, high-frequency rhythmic behavior during active feeding**. Typical reported values cluster around **150–180 contractions/min** during active feeding on yeast or standard media, and feeding rate **drops sharply** during the wandering/pre-pupal transition. Developmental comparisons are **rarely systematic across all instars**, but authors note that rates are lower and more variable in early instars, relatively stable during most of L3, and actively suppressed near pupation.

Stage	Reported rate	Notes	Source
late L2 / early L3	~ 160–$170\,\mathrm{min}^{-1}$	Active feeding on yeast	[1]
mid L3	~ 150–$180\,\mathrm{min}^{-1}$	Depends on hunger state	[6]
late L3	$\downarrow$ strongly	Feeding actively suppressed	[4]

Table 1: Quantitative anchor values for mouth-hook contraction rates in feeding assays.

Classical behavioral genetics & developmental feeding studies Earlier quantitative work explicitly examined **feeding rate vs larval age**, genetic variation in feeding behavior (e.g. rover/sitter phenotypes), and mouth-hook or cephalopharyngeal retraction counts as a *direct metric*.

Developmental phase	Feeding rate trend	Interpretation	Source
Early larva (L1)	Low, rising	Small mouthparts, immature circuit	[2, 7]
L2	Increasing	Growth acceleration	[2, 7]
Mid L3	Plateau	Sustained biomass accumulation	[2, 7]
Wandering / pre-pupa	Decline	Feeding inhibition precedes metamorphosis	[2, 3, 7]

Table 2: Developmental changes in larval feeding rate and interpretation.

Synthesis, implications, and bottom line

The combined modern and classical literatures agree that mouth-hook contraction rate is a **valid and widely used proxy** for feeding/bite frequency. Rates are **developmentally regulated**, not constant across instars: they rise early, plateau during much of L3, and are **actively suppressed** near pupation, with a **plateau during most of L3** emerging as a particularly strong empirical feature across studies.

Key gaps remain, including the lack of a **single standardized dataset** measuring mouth-hook frequency continuously from L1 -¿ L2 -¿ L3 under identical conditions, and the absence of direct, joint measurement of bite **frequency**, bite **duration**, and bite **volume** within the same experiments.

For DEB-style bite quantization, these findings imply that **constant bite frequency across the full larval period is empirically false**, although within L3 a roughly constant frequency is a reasonable approximation. Growth-related changes in assimilation are therefore most naturally attributed to changes in **bite volume** and in the **fraction of time spent feeding**, rather than to unbounded increases in frequency. Mouth-hook frequency can be treated as an *upper bound / rhythm*, with effective feeding rate given by frequency × bite volume × feeding duty cycle.

Taken together, these points make bite-quantized DEB models *biologically plausible* and highlight exactly where new data (continuous developmental time series and joint kinematic measurements) would be most informative.

Appendix: Rod-shape isomorph choice (shape correction included)

For your **rod-shaped isomorph** with a **shape correction** δ_M, a convenient mapping is to retain DEB's length-based scaling but correct for shape. The structural volume is

$$V(t) = (\delta_M L(t))^3 \qquad [\mathrm{cm}^3] \tag{22}$$

and the assimilation area (surface scaling) is

$$S_A(t) = (\delta_M L(t))^2 \qquad [\mathrm{cm}^2] \tag{23}$$

and thus

$$\dot{p}_A(t) = f(t)\,\{\dot{p}_{Am}\}\,(\delta_M L(t))^2. \tag{24}$$

So the bite frequency becomes

$$\boxed{\dot{N}_{\mathrm{bite}}(t) = \frac{(\delta_M L(t))^2}{V_{\mathrm{bite}}(t)}\,\frac{f(t)\,\{\dot{p}_{Am}\}}{\kappa_X\,\mu_X\,X(t)}}. \tag{25}$$

Note: if you later choose to use an explicit rod lateral area formula (rather than the $(\delta_M L)^2$ isomorph mapping), you can swap $S_A(t)$ accordingly without changing the rest of the derivation.

References

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